AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# MST129
# Applied Calculus

## Midterm Examination (MTA)
## Fall Semester 2025/2026

## Date: 17/Nov. /2025

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| **Number of Exam Pages: 3** (including this cover sheet) | **Time Allowed: 90 minutes** |

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**Instructions:**

- Total Marks: **60**
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.



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## Q–1: [5×3 marks] choose the correct answer:

**1.1:** The equation of the line with slope -3 and y-intercept 2 is :

**a)** Y= - 3x + 2
**b)** $Y = 3x - 2$
**c)** Y= -3x - 6
**d)** Y= 3x - 6
**e)** None of the above

**1.2:** The Derivative of $f(x) = (x^2 + 1)^{\frac{1}{2}}$ at x= -1 is:

**a)** $\frac{1}{\sqrt{2}}$
**b)** $\frac{-1}{\sqrt{2}}$
**c)** $\sqrt{2}$
**d)** $-\sqrt{2}$
**e)** None of the above

**1.3:** The slope of the tangent line to the curve of f(x)= $\frac{1}{x^3-4x+2}$ at $x = 1$ is:

**a)** 1/9
**b)** -1/9
**c)** 1/3
**d)** -1/3
**e)** None of the above

**1.4 :** The function $f(x) = x^4 + 1 has\ critical\ numher(s)\ at\ x =$ ?

**a)** 4
**b)** 5
**c)** 0
**d)** 1
**e)** None of the above

**1.5:** The (instantaneous) rate of change of f(x)= $\frac{-6}{x^2}$, $when\ x = 3$ is:

a) 2/9
b) 9/2
c) -2/3
d) -2/9
e) None of the above

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**Each question is worth 15 marks. Answer the following questions:**

**Q–2** Let a) $f(x) = x^2 + 2x + 1$ , and g(x)= $\sqrt{x-1}$

**i) [4 marks]** Find the domain of $f(x)$ and $g(x)$ in interval notation.

**ii) [5 marks]** Find $(g \circ f)(x)$.

b) **[6 marks]** Find the x- coordinate of the points of intersection of the curve $y=5x^2$ -3x-2 and y=2x-1

**Q–3:**

a) **Find** the slope of the tangent line to curve y= 4 $x^3 - 5x^2$ +2x +3 at (1,4) then write the equation of this tangent line.

b) **[8 marks] Find** the points on the graph of y= $\frac{8}{x^3-3x+1}$ at which the tangent line is horizontal.

**Q–4:** Let $f(x) = x^4 - 8x^2$

a) **[4 marks]** Find $f'(x)$ and $f''(x)$.

b) **[11 marks]** Find the intervals on which $f$ is increasing or decreasing and find the local maximum and minimum values of $f(x)$, if exist, by using first derivative test.

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**End of questions**



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